

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 3

Data Preparation

Module 9 Introduction

Introduction to Module 9's Study Video • 2 min

Why Data Preparation Is Not Optional in Machine Learning Reading • 5 min

Module 9-1 Basic Data Cleaning Review

Basic Data Cleaning Review Video • 38 sec

Materials on Basic Data Cleaning Reading • 10 min

Cleaning Data Is a Decision, Not a Checklist Reading • 5 min

Module 9-2 Distributions, Outliers and Their Removal

Distributions, Outliers and Their Removal: Gaussian Distribution, Skewness and Outliers Video • 6 min

Distributions, Outliers and Their Removal: The Z-Score and IQR Method for Outliers Removal Video • 5 min

Materials on Distributions, Outliers and Their Removal Reading • 10 min

Outliers, Assumptions, and When “Cleaning” Becomes Damage Reading • 5 min

Module 9-3 Data Transform

Data Transform: Normalization, Standardization, Power Transform Video • 7 min

Materials on Data Transform Reading • 10 min

When Scaling Helps — and When It Quietly Breaks Your Model Reading • 5 min

Module 9-4 Construct Training and Test Sets for Model Evaluation

Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling Video • 5 min

Construct Training and Test Sets for Model Evaluation: Cross-Validation Video • 6 min

Materials on Training/Test Sets Generation Reading • 10 min

Evaluation Starts with the Split: When Performance Numbers Lie Reading • 5 min

Module 9-5 Data Preparation Overview

Data Preparation Overview Video • 3 min

Materials on Data Preparation Reading • 10 min

From Raw Data to Reliable Learning: What Actually Matters? Reading • 5 min

Data Preparation and Evaluation Coach Dialogue • 5 min

Module 9 Quiz

Module 9 Quiz Practice Assignment • 30 min

Module 9 Assignments

Module 9 Assignment 1 Graded App Item • 30 min

Module 9 Assignment 2 Graded App Item • 30 min

Module 9 Assignment 3 Graded App Item • 30 min

XLS file for Module 9 Assignment Reading • 1 min

Module 9 Recap Reading • 5 min

Module 4

Support Vector Machines

Module 10 Introduction

Introduction to Module 10's Study Video • 1 min

Why Support Vector Machines? Reading • 5 min

Module 10-1 Support Vector Machine Fundamentals

Support Vector Machine Fundamentals: Concepts

When Scaling Helps — and When It Quietly Breaks Your Model



1. Reflecting on Why Scaling Matters
- Reflect on why features with very different numerical scales can distort the behavior of distance-based machine learning models such as k-nearest neighbors (kNN).
 - Consider an example where one feature could dominate the learning process simply because of its units or numerical range rather than its actual importance.
2. Reflecting on Choosing the Right Transformation
- Reflect on situations where normalization may be more appropriate than standardization.
 - Consider why standardization is often well-suited for approximately Gaussian-distributed data.
 - Reflect on the type of problem that power transformations, such as the Box–Cox transform, are designed to address beyond simple scaling techniques.
3. Reflecting on the Assumptions Behind Transformations
- min–max normalization,
 - standardization, and
 - power transformations such as Box–Cox.
 - Consider what problems or misleading effects could arise if these assumptions do not match the true characteristics of the data.
4. Reflecting on the Impact of Transformations on Models
- distance calculations in machine learning algorithms,
 - optimization and convergence during model training, and
 - the interpretation of learned parameters or predictions.
 - Consider why inverse transformations may sometimes be necessary when interpreting or presenting model outputs.
5. Reflecting on Connections to Future Topics
- outlier detection and handling,
 - train/test splitting, and
 - cross-validation.
 - Consider why transformations should be applied carefully to avoid introducing data leakage or misleading evaluation results.

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